

Hw 6

The intention of these problems is for you to work through the math yourself based on material covered in lecture, so that you can build mastery and understanding of the concepts. You should not use generative AI or seek out online resources that derive the same or similar problems. You may use (and should take advantage of) any functions in numpy and scipy. Please submit your work as a PDF on Gradescope.

1. An exponential random variable with parameter λ is specified by the pdf

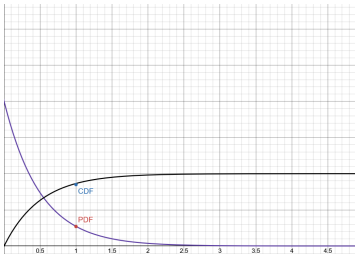
$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

This is often expressed compactly as $X \sim \text{Exp}(\lambda)$.

- a) Compute $F_X(x)$

$$\begin{aligned} F_X(x) &= \int_{-\infty}^x \lambda e^{-\lambda x'} dx' \\ &= \int_0^x \lambda e^{-\lambda x'} dx' \\ &= -e^{-\lambda x'} \Big|_0^x \\ &= 1 - e^{-\lambda x} \end{aligned}$$

- b) Draw $f_X(x)$ and $F_X(x)$



- c) Calculate $E(X)$

$$\frac{d}{dx} UV = UV' + VU'$$

Assuming that I'm finding

$$\begin{aligned} E(\lambda e^{-\lambda x}) &= \int_0^{\infty} x \lambda e^{-\lambda x} dx \\ &= \left(-x e^{-\lambda x} - \frac{1}{\lambda} e^{-\lambda x} \right) \Big|_0^{\infty} \\ &= \frac{1}{\lambda} \end{aligned}$$

- d) Calculate $\text{Var}(X)$

$$\text{Var}(X) = E[x^2] - E[X]^2$$

$$\begin{aligned} E[x^2] &= \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx \\ &= -x^2 e^{-\lambda x} - \frac{2x}{\lambda} e^{-\lambda x} - \frac{2}{\lambda^2} e^{-\lambda x} \Big|_0^{\infty} \\ &= \frac{2}{\lambda^2} \end{aligned}$$

So we have

$$\text{Var}[x] = \frac{1}{\lambda^2}$$

e) Let $X_1 \sim \text{Exp}(\lambda)$ and $X_2 \sim \text{Exp}(\lambda)$ be two independent random variables. Calculate ρ_{YZ} , where $Y = X_1 - X_2$ and $Z = X_1 + 2X_2$.

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

$$\begin{aligned}\sigma_{xy} &= E[(x - E[x])(Y - E[Y])] \\ \sigma_x^2 &= E[x^2] - E[x]^2\end{aligned}$$

$$\sigma_z$$

Because X_1 and X_2 are independent,

$$\begin{aligned}\text{Var}[Z] &= \text{Var}[X_1 + 2X_2] \\ &= \text{Var}[X_1] + \text{Var}[2X_2] \\ &= \frac{1}{\lambda^2} + \frac{4}{\lambda^2} \\ &= \frac{5}{\lambda^2}\end{aligned}$$

$\text{Var}[x]$ is nonlinear, so constant terms inside the argument pull out as squares.

This gives us that

$$\sigma_z = \frac{\sqrt{5}}{\lambda}$$

$$\sigma_y$$

$$\begin{aligned}\text{Var}[Y] &= \text{Var}[X_1] + \text{Var}[X_2] \\ &= \frac{2}{\lambda^2}\end{aligned}$$

$$\sigma_{yz}$$

$$\begin{aligned}\sigma_{zy} &= E[(Z - E[Z])(Y - E[Y])] \\ \sigma_{zy} &= E[ZY] - E[Z]E[Y]\end{aligned}$$

Also,

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_x(x) dx$$

where

$$\begin{aligned}g(x) &= ZY \\ &= (X_1 - X_2)(X_1 + 2X_2) \\ &= X_1^2 + 2X_2X_1 - X_2X_1 - 2X_2^2 \\ &= X_1^2 + X_1X_2 - 2X_2^2\end{aligned}$$

We can now find

$$\begin{aligned}E[ZY] &= E[X_1^2] + E[X_1]E[X_2] - 2E[X_2^2] \\ &= \frac{2}{\lambda^2} + \frac{1}{\lambda^2} - 2\frac{2}{\lambda^2} \\ &= -\frac{1}{\lambda^2}\end{aligned}$$

Now we need

$$E[Z]E[Y]$$

$$\begin{aligned} E[Y] &= E[X_1 - X_2] \\ &= 0 \end{aligned}$$

So this is just zero.

$$\sigma_{yz} = -\frac{1}{\lambda^2}$$

ρ_{yz}

$$\begin{aligned} \rho_{yz} &= \frac{\sigma_{yz}}{\sigma_z \sigma_y} \\ &= \frac{\left(-\frac{1}{\lambda^2}\right)}{\frac{2}{\lambda^2} \frac{\sqrt{5}}{\lambda}} \\ &= \frac{-1}{\lambda^2} \frac{\lambda^3}{2\sqrt{5}} \\ &= -\frac{\lambda}{2\sqrt{5}} \end{aligned}$$

2. A random variable Z is determined in the following way. First, a coin is flipped. If the coin is heads, then a number is drawn from an exponential distribution with parameter λ . If the coin flip is tails, then a number is drawn from a uniform distribution in the interval $[a, b]$, where $0 < a < b$.

- a) Find an analytical expression for $f_Z(z)$. You may express your answer in terms of the step function

$$u(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

If we label heads as $+1$ and tails as -1 , then can use the step as it is

$$f_Z(z) = 0.5 \left(\frac{1}{b-a} (u[x-a] - u[x-b]) + \lambda e^{-\lambda x} \right)$$

- b) Find an analytical expression for $F_Z(z)$.

$$F_Z(z) = \int_{-\infty}^{\infty} f_Z(z)$$

We have three regions - before the uniform distribution, during it up to the end, and after it. I'll use X because I accidentally did all the work with x , and I'm too lazy to change it.

$$= \begin{cases} 0.5 \left(\int_0^x \lambda e^{-\lambda x} dx + 0 \right) & x < a \\ 0.5 \left(\int_0^a \lambda e^{-\lambda x} dx + \int_a^x \frac{1}{b-a} + \lambda e^{-\lambda x} \right) dx & a < x < b \\ 0.5 \left(\int_0^a \lambda e^{-\lambda x} dx + \int_a^b \frac{1}{b-a} + \lambda e^{-\lambda x} dx + \int_b^x \lambda e^{-\lambda x} dx \right) & b < x \end{cases}$$

case 1

$$\begin{aligned} &\int_0^x 0.5 \lambda e^{-\lambda x'} dx' \\ &= -0.5 e^{-\lambda x'} \Big|_0^x \\ &= 0.5 (1 - e^{-\lambda x}) \end{aligned}$$

Case 2

$$\begin{aligned}
& 0.5 \left(1 - e^{-\lambda a} + \int_a^x \frac{1}{b-a} + \lambda e^{-\lambda x'} dx' \right) \\
&= 0.5 \left(1 - e^{-\lambda a} + \frac{1}{b-a} x' \Big|_a^x - e^{-\lambda x'} \Big|_a^x \right) \\
&= 0.5 \left(1 - e^{-\lambda a} + \frac{x-a}{b-a} - e^{-\lambda x} + e^{-\lambda a} \right) \\
&= 0.5 \left(1 + \frac{x-a}{b-a} - e^{-\lambda x} \right)
\end{aligned}$$

Case 3

$$\begin{aligned}
&= 0.5 \left(\underbrace{1}_{\text{uniform}} + 1 - e^{-\lambda x} \right) \\
&= 0.5(2 - e^{-\lambda x})
\end{aligned}$$

This gives us

$$F_Z(z) = \begin{cases} 0.5(1 - e^{-\lambda z}) & z < a \\ 0.5 \left(1 + \frac{z-a}{b-a} - e^{-\lambda z} \right) & a < z < b \\ 0.5(2 - e^{-\lambda z}) & b < z \end{cases}$$

c) Compute $E(Z)$

$$\begin{aligned}
E[z] &= \int_{-\infty}^{\infty} z f_z(z) dz \\
f_Z(z) &= 0.5 \left(\frac{1}{b-a} (u[x-a] - u[x-b]) + \lambda e^{-\lambda x} \right)
\end{aligned}$$

We can just separate this out - the uniform distribution just goes to its average value, and we found for the exponential

$$\begin{aligned}
& 0.5 * \int_a^b \frac{1}{b-a} x dx + 0.5 * \int_0^{\infty} x \lambda e^{-\lambda x} dx \\
&= \frac{1}{2} \left(\frac{a+b}{2} + \frac{1}{\lambda} \right)
\end{aligned}$$

d) Compute $Var(Z)$

$$\begin{aligned}
Var[z] &= 0.5 * \int_a^b \frac{1}{b-a} x^2 dx + 0.5 * \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx \\
&= \frac{1}{2} \left(\frac{b^3 - a^3}{3(b-a)} + \frac{1}{\lambda^2} \right) \\
&= \frac{1}{2} \left(\frac{a^2 + ab + b^2}{3} + \frac{1}{\lambda^2} \right)
\end{aligned}$$

(I looked up the $b^3 - a^3$ expansion)

e) Calculate ρ_{z_1, z_2} , the correlation coefficient between two independent draws from the distribution.

$$\sigma_{z_1 z_2} = E[z_1 z_2] - E[z_1]E[z_2]$$

Because we are dealing with the raw draws and not complicated functions built off of the draws, there is no correlation -

$$E[z_1 z_2] = E[z_1]E[z_2]$$

so $\sigma_{z_1 z_2} = 0$

Therefore, $\rho_{z_1, z_2} = 0$

3. Recall from class that a maximum likelihood model is the one that maximizes the conditional pdf $f(x|\theta)$.
- a) You draw n independent samples $x_1, x_2, \dots, x_n$ from an exponential distribution with unknown parameter λ . Calculate the maximum likelihood estimate of λ .

The chance of all draws is

$$\prod_{i=1}^n \lambda e^{-\lambda x_i}$$

$$\lambda^n \prod_{i=1}^n e^{-\lambda x_i}$$

We can find λ by minimizing the product

$$\frac{d}{d\lambda} \lambda^n \prod_{i=1}^n e^{-\lambda x_i} = 0$$

We can simplify with the natural log, since multiplication of things with exponents is addition of the exponents, and ln pulls apart multiplication as addition, so this becomes

$$\frac{d}{d\lambda} \ln(\lambda^n) + \sum_{i=1}^n -\lambda x_i = 0$$

$$\frac{d}{d\lambda} \ln(\lambda^n) - \lambda \sum_{i=1}^n x_i = 0$$

$$\frac{n}{\lambda} - \sum_{i=1}^n x_i = 0$$

$$\lambda = \frac{n}{\sum_{i=1}^n x_i}$$

- b) You draw n independent samples $x_1, x_2, \dots, x_n$ from a Gaussian distribution with unknown mean μ and unknown variance σ^2 . Calculate the maximum likelihood estimates of μ and σ^2 .

$$f_X(x, \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

We can do the same process as before. Lets start with the means

$$\frac{1}{\sqrt{2\pi\sigma^2}} \prod_{i=1}^n e^{-\frac{1}{2\sigma^2}(x_i-\mu)^2}$$

We want to minimize this. Constants don't matter, we care about what minimizes

$$\prod_{i=1}^n e^{-(x_i-\mu)^2}$$

So

$$\frac{d}{d\mu} \prod_{i=1}^n e^{-(x_i-\mu)^2} = 0$$

$$\frac{d}{d\mu} \sum_{i=1}^n (x_i - \mu)^2 = 0$$

$$\frac{d}{d\mu} \sum_{i=1}^n x_i^2 + \mu^2 - 2x_i\mu = 0$$

$$\sum_{i=1}^n 2\mu - 2x_i = 0$$

$$\sum_{i=1}^n x_i = n\mu$$

$$\mu = \frac{\sum_{i=1}^n x_i}{n}$$

This makes sense as the definition of the mean.

Lets do it for the standard deviation now:

$$\frac{1}{\sqrt{2\pi\sigma^2}} \prod_{i=1}^n e^{-\frac{1}{2\sigma^2}(x_i - \mu)^2}$$

We can't chuck out the constants, but we still have some manipulation with the ln

$$\begin{aligned} \frac{d}{d\sigma} \left(-n \ln \sqrt{2\pi\sigma^2} + \sum_{i=1}^n -\frac{1}{2\sigma^2}(x_i - \mu)^2 \right) &= 0 \\ \frac{d}{d\sigma} n \ln \sigma - \ln \sqrt{2\pi} - \frac{1}{\sigma^3} \sum_{i=1}^n (x_i - \mu)^2 &= 0 \\ \frac{n}{\sigma} - \frac{1}{\sigma^3} \sum_{i=1}^n (x_i - \mu)^2 &= 0 \\ \sigma^2 &= \frac{\sum_{i=1}^n (x_i - \mu)^2}{n} \end{aligned}$$

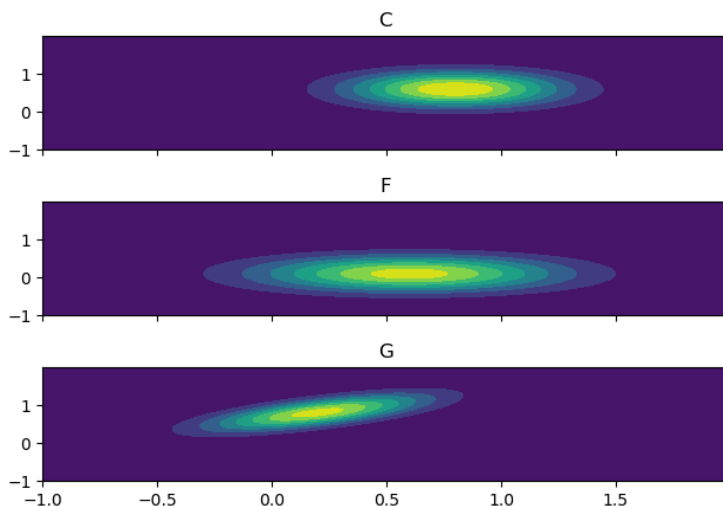
This is the definition of the variance, which also makes sense.

4. You would like to estimate the chord in a given audio frame. The three chord models for C major, F major, and G major chords are 2-dimensional Gaussians with $\mu_C = [.8 \ .6]^T$, $\Sigma_C = \begin{pmatrix} .1 & 0 \\ 0 & .1 \end{pmatrix}$, $\mu_F = [.6 \ .1]^T$, $\Sigma_F = \begin{pmatrix} .2 & 0 \\ 0 & .1 \end{pmatrix}$, $\mu_G = [.2 \ .8]^T$, $\Sigma_G = \begin{pmatrix} .1 & .07 \\ .07 & .1 \end{pmatrix}$. The first dimension corresponds to the energy in the C chroma, and the second dimension corresponds to energy in the G chroma. (Using chroma features we might want to use a 12 dimensional space, but for simplicity we will only consider a 2 dimensional space in this problem. Similarly, we might want to consider all possible major and minor chords, but we will only consider these 3 chords in this problem.) Let S be the random variable indicating which of these three chords the current frame is in, and let x be the 2-dimensional vector representing the C and G chroma energies in the current frame.

Please answer the following questions below in a jupyter notebook, run the notebook so all outputs and plots are showing, convert it to PDF format, and include it as part of your homework submission.

- a) Plot $f_{X|S}(x|C \text{ major})$, $f_{X|S}(x|F \text{ major})$, and $f_{X|S}(x|G \text{ major})$. You may use the function `scipy.stats.multivariate_normal`.

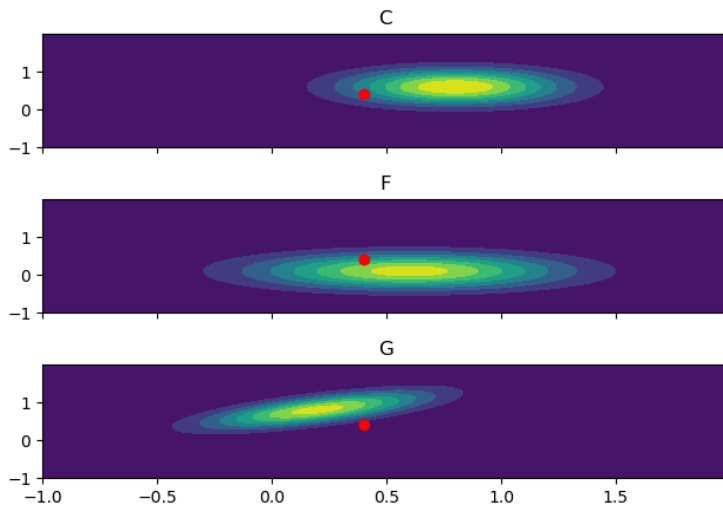
Multivariate gaussians for chords



- b) Given that $X = [.4 \ .4]^T$, estimate which of the three chords is most likely using the maximum likelihood criterion.

This looks to be the fewest deviations away from the F chord, so most likely from that.

Multivariate gaussians for chords

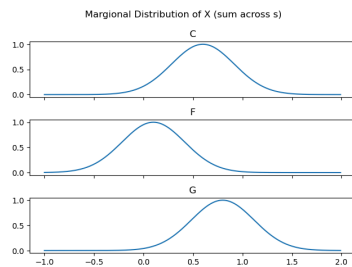


- c) Now assume that you know that the previous audio frame was a C major chord. Given this information, the probability that the current audio frame is a C major, F major, and G major chords is .4, .3, and .3, respectively. Determine which model maximizes $f_{X,S}(x,s)$. This is called the maximum a posteriori (MAP) estimate. The MAP estimate takes into account prior beliefs into our estimate of the best model. Note that maximum likelihood is a special case of MAP where all models have equal priors.

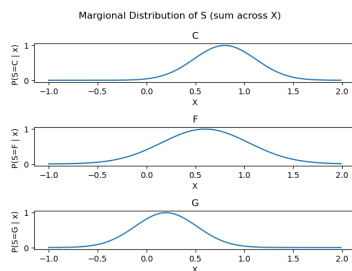
The probabilities in order of C, F, G are

0.234, 0.195, 0.0314. The model for C maximizes the probability, even though F is closer to the center.

- d) Plot $f_X(x)$, the marginal distribution of X .



- e) Calculate $P(S = C|x)$, $P(S = F|x)$, and $P(S = G|x)$. What is the connection of these quantities to MAP?



Given that the previous was a c,
C 0.4 F 0.3 G 0.3

This feels like the MAP values are the distance from the means to each peak of the amount of overlap if I integrate $C * G$ vs $C * F$.

5. Please indicate the number of hours you spent on this assignment: _____